

3/9/23

Chemistry

1 BaSO_4 , White

2

3 Both A and B are true, R is correct explanation of A

4 A is True, R is false

5 Amphoteric oxides are metal oxides that react with acids and bases to form salts and water.

6 Only in case of Aluminium and zinc would the student observe displacement reaction. This is because for a displacement reaction to take place, a more reactive metal is needed to displace a less reactive metal. Here, only Aluminium and Zinc are more reactive than Iron.

7 $2\text{FeSO}_4 \xrightarrow{\Delta} \text{Fe}_2\text{O}_3 + \text{SO}_2 \uparrow + \text{SO}_3 \uparrow$
 Color of solid product is Reddish Brown

8

a Zinc when reacts with dil. HCl forms bubbles of H_2 gas on its surface.
 $\text{Zn} + 2\text{HCl}_{(\text{dil})} \rightarrow \text{ZnCl}_2 + \text{H}_2$ and ZnCl_2 is formed

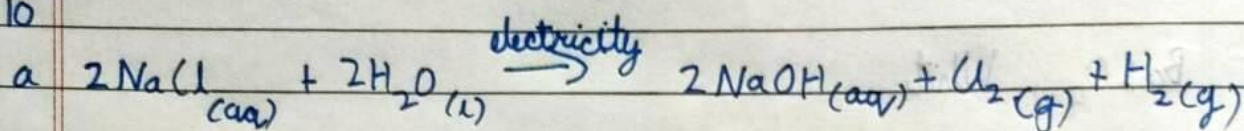
b Egg shell has CaCO_3 as its main constituent. When we add dil. HCl the CaCO_3 reacts and evolves CO_2 . The egg shell slowly dissolves.
 $\text{CaCO}_3 + 2\text{HCl}_{(\text{dil})} \rightarrow \text{CaCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$

9 Salt is Baking Soda (NaHCO_3)

It is added for cooking faster

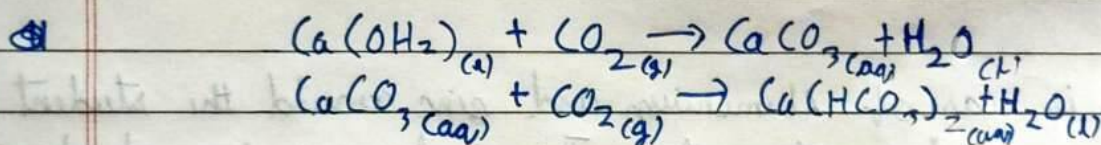
It can be used in soda acid fire extinguisher

10



This process is called Chlor-Alkali process

- b On passing CO_2 gas through lime water (Ca(OH)_2) it turns milky white due to formation of insoluble Calcium carbonate (CaCO_3). But Excess CO_2 turns this solution colourless as the CaCO_3 reacts with excess CO_2 to form $\text{Ca(HCO}_3)_2$ which is soluble and dissolves to make the solution colourless.



11 A = $\text{Pb(NO}_3)_2$ Lead nitrate

B = NO_2 Nitrogen Dioxide

C = O_2 Oxygen

D = PbI_2 Lead Iodide

First reaction - $2\text{Pb(NO}_3)_2 \xrightarrow{\Delta} 2\text{PbO} + 4\text{NO}_2 + \text{O}_2$ and is decomposition reaction

Second reaction - $\text{PbO} + 2\text{KI} \rightarrow \text{PbI}_2 + \text{K}_2\text{O}$ it is Double displacement reaction